PGCP-AC AUGUST 26
Database Assignment 1

Schema Design & Table Creation (DDL & DML Commands)

1. Define a schema for a Library Management System with the following entities:
   - Authors
   - Books
   - Members
   - Borrow_Records

```
mysql> create database library_management_system;
Query OK, 1 row affected (0.01 sec)

mysql> use library_management_system;
Database changed
```

2. Write the SQL command to create a table **Authors** with the following fields:
   - author_id (Primary Key, INT)
   - name (VARCHAR(100))
   - country (VARCHAR(50))

```
mysql> create table author(
    -> author_id int primary key,
    -> name VARCHAR(100),
    -> country VARCHAR(50));
Query OK, 0 rows affected (0.06 sec)

mysql> show author;
ERROR 1064 (42000): You have an error in your SQL syntax; check t
uthor' at line 1
mysql> desc author;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| author_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(100) | YES  |     | NULL    |       |
| country   | varchar(50)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
3 rows in set (0.03 sec)
```

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3. Write the SQL command to create a table **Books** with the following fields:

- book_id (Primary Key, INT)
- title (VARCHAR(150))
- author_id (Foreign Key referencing Authors)
- published_year (YEAR)

```
mysql> create table books(
    -> book_id INT Primary Key,
    -> title VARCHAR(150),
    -> author_id int,
    -> published_year YEAR,
    -> available_copies INT,
    -> Foreign Key(author_id) references Author(author_id));
Query OK, 0 rows affected (0.04 sec)

mysql> show books;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manu
ooks' at line 1
mysql> desc books;
+------------------+--------------+------+-----+---------+-------+
| Field            | Type         | Null | Key | Default | Extra |
+------------------+--------------+------+-----+---------+-------+
| book_id          | int          | NO   | PRI | NULL    |       |
| title            | varchar(150) | YES  |     | NULL    |       |
| author_id        | int          | YES  | MUL | NULL    |       |
| published_year   | year         | YES  |     | NULL    |       |
| available_copies | int          | YES  |     | NULL    |       |
+------------------+--------------+------+-----+---------+-------+
5 rows in set (0.02 sec)
```

4. Write the SQL command to create a table **Members** with:

- member_id (Primary Key, INT)
- name (VARCHAR(100))
- email (VARCHAR(100), unique)
- phone (VARCHAR(15))

```
mysql> create table members(
    -> member_id INT Primary Key,
    -> name VARCHAR(50),
    -> email VARCHAR(100) UNIQUE,
    -> phone VARCHAR(15));
Query OK, 0 rows affected (0.03 sec)

mysql> DESC MEMBERS;
+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| member_id | int          | NO   | PRI | NULL    |       |
| name      | varchar(50)  | YES  |     | NULL    |       |
| email     | varchar(100) | YES  | UNI | NULL    |       |
| phone     | varchar(15)  | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

5. Write the SQL command to create a table **Borrow_Records** with:
   - record_id (Primary Key, INT)
   - member_id (Foreign Key referencing Members)
   - book_id (Foreign Key referencing Books)
   - borrow_date (DATE)
   - return_date (DATE)

```
mysql> create table Borrow_Records(
    -> record_id INT Primary Key,
    -> member_id INT,
    -> book_id INT,
    -> borrow_date DATE,
    -> return_date DATE,
    -> Foreign Key(member_id) references Members(member_id),
    -> Foreign Key(book_id) references Books(book_id));
Query OK, 0 rows affected (0.09 sec)

mysql> desc Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.03 sec)
```

6. Modify the Books table to add a column genre of type VARCHAR(50).

```
mysql> alter table books
    -> add column type VARCHAR(50);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc books;
+-----------------+--------------+------+-----+---------+-------+
| Field           | Type         | Null | Key | Default | Extra |
+-----------------+--------------+------+-----+---------+-------+
| book_id         | int          | NO   | PRI | NULL    |       |
| title           | varchar(150) | YES  |     | NULL    |       |
| author_id       | int          | YES  | MUL | NULL    |       |
| published_year  | year         | YES  |     | NULL    |       |
| available_copies | int         | YES  |     | NULL    |       |
| type            | varchar(50)  | YES  |     | NULL    |       |
+-----------------+--------------+------+-----+---------+-------+
6 rows in set (0.01 sec)
```

7. Write the SQL command to drop the Borrow_Records table.

```
mysql> drop table Borrow_Records;
Query OK, 0 rows affected (0.05 sec)

mysql> desc Borrow_Records;
ERROR 1146 (42S02): Table 'library_management_system.borrow_records' doesn't exist
```

8. Insert 3 records into the Authors table.

```
mysql> insert into author(author_id, name, country)
    -> values
    -> (101, 'Alice', 'Swizerland'),
    -> (102, 'Bob', 'France'),
    -> (103, 'Casey', 'Japan');
Query OK, 3 rows affected (0.04 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from author;
+-----------+-------+------------+
| author_id | name  | country    |
+-----------+-------+------------+
|       101 | Alice | Swizerland |
|       102 | Bob   | France     |
|       103 | Casey | Japan      |
+-----------+-------+------------+
3 rows in set (0.02 sec)
```

9. Insert 5 books into the Books table.

```
mysql> insert into books(book_id, title, author_id, published_year, available_copies, type)
    -> values
    -> (201, 'Harry Potter', 103, '1997', 37, 'Magical'),
    -> (202, 'Spider-man', 102, '1990', 28, 'SuperHero'),
    -> (203, 'Love in tokyo', 101, '1984', 48, 'Romance'),
    -> (204, 'Super-man', 102, '1981', 21, 'SuperHero'),
    -> (205, 'Lord of rings', 103, '1995', 25, 'Magical');
Query OK, 5 rows affected (0.03 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> select * from books;
+---------+--------------+-----------+----------------+------------------+-----------+
| book_id | title        | author_id | published_year | available_copies | type      |
+---------+--------------+-----------+----------------+------------------+-----------+
|     201 | Harry Potter |       103 |           1997 |               37 | Magical   |
|     202 | Spider-man   |       102 |           1990 |               28 | SuperHero |
|     203 | Love in tokyo |       101 |           1984 |               48 | Romance   |
|     204 | Super-man    |       102 |           1981 |               21 | SuperHero |
|     205 | Lord of rings |       103 |           1995 |               25 | Magical   |
+---------+--------------+-----------+----------------+------------------+-----------+
5 rows in set (0.00 sec)
```

10. Insert 3 members into the Members table.

```
mysql> insert into members (member_id, name, email, phone)
    -> values
    -> (301, 'Steve', 'steve@gmail.com', '2313435678'),
    -> (302, 'Nancy', 'nancy@gmail.com', '8291731930'),
    -> (303, 'John', 'john@gmail.com', '4628781372');
Query OK, 3 rows affected (0.03 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from members;
+-----------+-------+-----------------+------------+
| member_id | name  | email           | phone      |
+-----------+-------+-----------------+------------+
|       301 | Steve | steve@gmail.com | 2313435678 |
|       302 | Nancy | nancy@gmail.com | 8291731930 |
|       303 | John  | john@gmail.com  | 4628781372 |
+-----------+-------+-----------------+------------+
3 rows in set (0.00 sec)
```

11. Insert 4 borrow records into the Borrow_Records table.

```
mysql> insert into Borrow_Records(record_id, member_id, book_id, borrow_date, return_date)
    -> values
    -> (401, 301, 203, '2025-03-12', '2025-03-19'),
    -> (402, 302, 205, '2025-11-24', '2025-11-30'),
    -> (403, 303, 202, '2026-02-08', '2026-02-16'),
    -> (404, 302, 201, '2026-04-03', '2026-04-10');
Query OK, 4 rows affected (0.03 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * from Borrow_Records;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       401 |       301 |     203 | 2025-03-12  | 2025-03-19  |
|       402 |       302 |     205 | 2025-11-24  | 2025-11-30  |
|       403 |       303 |     202 | 2026-02-08  | 2026-02-16  |
|       404 |       302 |     201 | 2026-04-03  | 2026-04-10  |
+-----------+-----------+---------+-------------+-------------+
4 rows in set (0.00 sec)
```

12. Write an SQL query to select all books where published_year is after 1990.

```
mysql> select * from books
    -> where published_year > '1990';
+---------+---------------+-----------+----------------+------------------+---------+
| book_id | title         | author_id | published_year | available_copies | type    |
+---------+---------------+-----------+----------------+------------------+---------+
|     201 | Harry Potter  |       103 |           1997 |               37 | Magical |
|     205 | Lord of rings |       103 |           1995 |               25 | Magical |
+---------+---------------+-----------+----------------+------------------+---------+
2 rows in set (0.01 sec)
```

13. Write a SQL query to create a foreign key & primary key relationship between two tables.

```
mysql> create table Borrow_Records(
    -> record_id INT Primary Key,
    -> member_id INT,
    -> book_id INT,
    -> borrow_date DATE,
    -> return_date DATE,
    -> Foreign Key(member_id) references Members(member_id),
    -> Foreign Key(book_id) references Books(book_id));
Query OK, 0 rows affected (0.09 sec)

mysql> desc Borrow_Records;
+-------------+------+------+-----+---------+-------+
| Field       | Type | Null | Key | Default | Extra |
+-------------+------+------+-----+---------+-------+
| record_id   | int  | NO   | PRI | NULL    |       |
| member_id   | int  | YES  | MUL | NULL    |       |
| book_id     | int  | YES  | MUL | NULL    |       |
| borrow_date | date | YES  |     | NULL    |       |
| return_date | date | YES  |     | NULL    |       |
+-------------+------+------+-----+---------+-------+
5 rows in set (0.03 sec)
```

14. Write an SQL query to find all members who have borrowed the book with title 'Harry Potter'.

```
mysql> select * from Borrow_Records
    -> where book_id = 201;
+-----------+-----------+---------+-------------+-------------+
| record_id | member_id | book_id | borrow_date | return_date |
+-----------+-----------+---------+-------------+-------------+
|       404 |       302 |     201 | 2026-04-03  | 2026-04-10  |
+-----------+-----------+---------+-------------+-------------+
1 row in set (0.02 sec)
```

15. Update the available_copies column of a specific book (choose any book) by reducing it by 1 after it is borrowed.

```
mysql> update books
    -> set available_copies = available_copies - 1
    -> where book_id = 201;
Query OK, 1 row affected (0.04 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from books;
+---------+--------------+-----------+----------------+------------------+-----------+
| book_id | title        | author_id | published_year | available_copies | type      |
+---------+--------------+-----------+----------------+------------------+-----------+
|     201 | Harry Potter |       103 |           1997 |               36 | Magical   |
|     202 | Spider-man   |       102 |           1990 |               28 | SuperHero |
|     203 | Love in tokyo |       101 |           1984 |               48 | Romance   |
|     204 | Super-man    |       102 |           1981 |               21 | SuperHero |
|     205 | Lord of rings |       103 |           1995 |               25 | Magical   |
+---------+--------------+-----------+----------------+------------------+-----------+
5 rows in set (0.00 sec)
```

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16. Delete a record from Members where member_id = 3.

```
mysql> delete from borrow_records
    -> where member_id = 303;
Query OK, 1 row affected (0.03 sec)

mysql> delete from members
    -> where member_id = 303;
Query OK, 1 row affected (0.02 sec)

mysql> select * from members;
+-----------+-------+-----------------+-------------+
| member_id | name  | email           | phone       |
+-----------+-------+-----------------+-------------+
|       301 | Steve | steve@gmail.com | 2313435678 |
|       302 | Nancy | nancy@gmail.com | 8291731930 |
+-----------+-------+-----------------+-------------+
2 rows in set (0.00 sec)
```

17. Update a Book name record from Book table with id = 204.

```
mysql> update books
    -> set title = 'Avengers'
    -> where book_id = 204;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from books;
+---------+--------------+-----------+----------------+------------------+-----------+
| book_id | title        | author_id | published_year | available_copies | type      |
+---------+--------------+-----------+----------------+------------------+-----------+
|     201 | Harry Potter |       103 |           1997 |               36 | Magical   |
|     202 | Spider-man   |       102 |           1990 |               28 | SuperHero |
|     203 | Love in tokyo |       101 |           1984 |               48 | Romance   |
|     204 | Avengers     |       102 |           1981 |               21 | SuperHero |
|     205 | Lord of rings |       103 |           1995 |               25 | Magical   |
+---------+--------------+-----------+----------------+------------------+-----------+
5 rows in set (0.00 sec)
```

18. Write an SQL query to list all books along with their authors' names.

```
mysql> SELECT b.title, a.name
    -> FROM Books b
    -> INNER JOIN Author a
    -> ON b.author_id = a.author_id;
+---------------+-------+
| title         | name  |
+---------------+-------+
| Love in tokyo | Alice |
| Spider-man    | Bob   |
| Avengers      | Bob   |
| Harry Potter  | Casey |
| Lord of rings | Casey |
+---------------+-------+
5 rows in set (0.00 sec)
```

19. Write an SQL query to delete all books from the Books table where the published_year is before 2000.

```
mysql> DELETE FROM borrow_records
    -> WHERE book_id IN (
    ->     SELECT book_id
    ->     FROM books
    ->     WHERE published_year < 1990
    -> );
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> DELETE FROM books
    -> WHERE published_year < 1990;
Query OK, 2 rows affected (0.02 sec)

mysql> select * from books;
+---------+---------------+-----------+----------------+------------------+-----------+
| book_id | title         | author_id | published_year | available_copies | type      |
+---------+---------------+-----------+----------------+------------------+-----------+
|     201 | Harry Potter  |       103 |           1997 |               36 | Magical   |
|     202 | Spider-man    |       102 |           1990 |               28 | SuperHero |
|     205 | Lord of rings |       103 |           1995 |               25 | Magical   |
+---------+---------------+-----------+----------------+------------------+-----------+
3 rows in set (0.00 sec)
```

20. Write an SQL query to find all books that are never borrowed (i.e., no records in Borrow_Records**)**.

```
mysql> SELECT b.*
    -> FROM Books b
    -> LEFT JOIN Borrow_Records br
    -> ON b.book_id = br.book_id
    -> WHERE br.book_id IS NULL;
+---------+------------+-----------+----------------+------------------+-----------+
| book_id | title      | author_id | published_year | available_copies | type      |
+---------+------------+-----------+----------------+------------------+-----------+
|     202 | Spider-man |       102 |           1990 |               28 | SuperHero |
+---------+------------+-----------+----------------+------------------+-----------+
1 row in set (0.01 sec)
```